

Applied A.I. Solutions

Agile Project Management Methodologies

Quotations, Reference and Recommended Bibliography:

- 1 - "The Epic Guide to Agile – More Business Value On A Predictable Schedule With Scrum" – Author: Dave Todaro - Copyright © 2019 David M. Todaro – Published by R9 Publishing LLC
- 2 - "Agile Project Management with Scrum" – Author: Ken Schwaber – Copyright © 2004 by Ken Schwaber – Published by Microsoft Press
- 3 - "Agile Project Management with Kanban" – Author: Eric Brechner – Copyright © 2015 by Eric Brechner – Published by Microsoft Press

Agile Project Management Methodologies

Ritual + Rhythm + Communication + Value +
Commitment + Teamwork + Discipline + Transparency
+ Continuous Improvement + Rapid Development

Week 2 / Day 1

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Kanban

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Kanban

- Kanban is a simple project management technique that is based on Toyota's just-in-time scheduling mechanism.
- Using Kanban to manage a project work will allow to focus all time and energy on delivering value to the customer.
- Kanban means ...
 - "signboard" in Japanese (not a perfect translation though)
 - "sign card" at Toyota; note cards that track and control work
 - "looking at the board", from Chinese, referring to the daily standup meeting

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Kanban characteristics

- Planning meetings only on demand and no special meetings about process.
- Kanban visualizes project workflow, spotlights bottlenecks the day they occur, and forces team members to immediately resolve the issue.
- Prevents careless team members from prematurely designating work as complete.

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Kanban characteristics – cont'd

- Enforces clear quality bars at each step, driving quality upstream.
- Minimizes work progress, freeing teams to adjust daily to new priorities and requirements with little sunk cost and allowing a team to deliver on time, on budget.

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Kanban quick-start guide

- Step 1: Capture your team's high-level routine
- Step 2: Redecorate your wall
- Step 3: Set limits on chaos
- Step 4: Define "done"
- Step 5: Run your daily standup
- Troubleshooting
- Checklist

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Kanban quick-start guide

- **Step 1:** Capture your team's high-level routine
 - Discuss the product with partners, teammates, and customers
 - Find, evaluate, and fix bugs and operational issues
 - Produce improvements to products and infrastructure

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Kanban quick-start guide

Step 1: Capture your team's high-level routine – cont'd

- ✓ Take an item from the backlog
 - ✓ Specify the work necessary to implement it
 - ✓ Implement it
 - ✓ Validate that it works properly
 - ✓ Deliver it to customers or partners



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Kanban quick-start guide

- **Step 2:** Redecorate your wall
 - Visualizing workflow is one of Kanban's core principles
 - Post the steps on a signboard close to where your team resides
 - The signboard will be used to visualize progress and attach note cards to it to track their work
 - Manipulate cards on the whiteboard
 - Use online tools to track the work too – useful for distributed teams

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Kanban quick-start guide

- **Step 2:** Redecorate your wall – cont’d

Backlog	Specify work (2)		Implement it (5)		Validate it (3)	
	Active	Completed	Active	Completed	Active	Completed
     	 		 		  	
signboard						

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Kanban quick-start guide

- **Step 3: Set limits on chaos**
 - Chaos is limited by directly limiting the amount of work in progress (WIP)
 - It limits the amount of work affected by changing priorities, requirements, or designs
 - It restricts the flow of work to match the pace of the slowest step (constraint)
 - It translates in the numbers of note cards allowed at each step

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Kanban quick-start guide

- **Step 3: Set limits on chaos – cont'd**
 - Determine WIP limits
 - Set the WIP limit for your slowest step to the number of team members doing that step, plus a 50% buffer
 - Use ratios to set the WIP limits for the other steps

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Kanban quick-start guide

- **Step 3: Set limits on chaos – cont'd**

- Determine WIP limits

on average

constraint

Step	Specify	Implement	Validate
A: Average rate per month per person	6	2	3
B: Slowest rate (min A:)		2	
C: Number of people assigned to step B		3	
D: Throughput of step B (B*C)		6	
E: People needed to match B's throughput (D:/A:)	1	3	2
F: WIP limits (E*1.5 rounded up)	2	5	3

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Kanban quick-start guide

- **Step 4: Define “done” rule**
 - Specify step: all items are broken down into tasks that can be finished in less than a week each, and quick specifications completed for each item.
 - Implement step: code is reviewed, and unit tested, the code is checked in, acceptance tests pass, and the user documentation is complete.
 - Validate step: the work is deployed to production and tried by a significant subset of real customers. All issues found are resolved.

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Kanban quick-start guide

- **Step 5: Run your Daily standup**
 - The only meetings are daily stand-ups at the team signboard
 - The only required question is asking whether any team members are blocked or need assistance
 - With a loaded backlog, no other meetings are required
 - There are no milestones, No sprints, No retrospective
 - Kanban flows continuously, so long as there is work to do

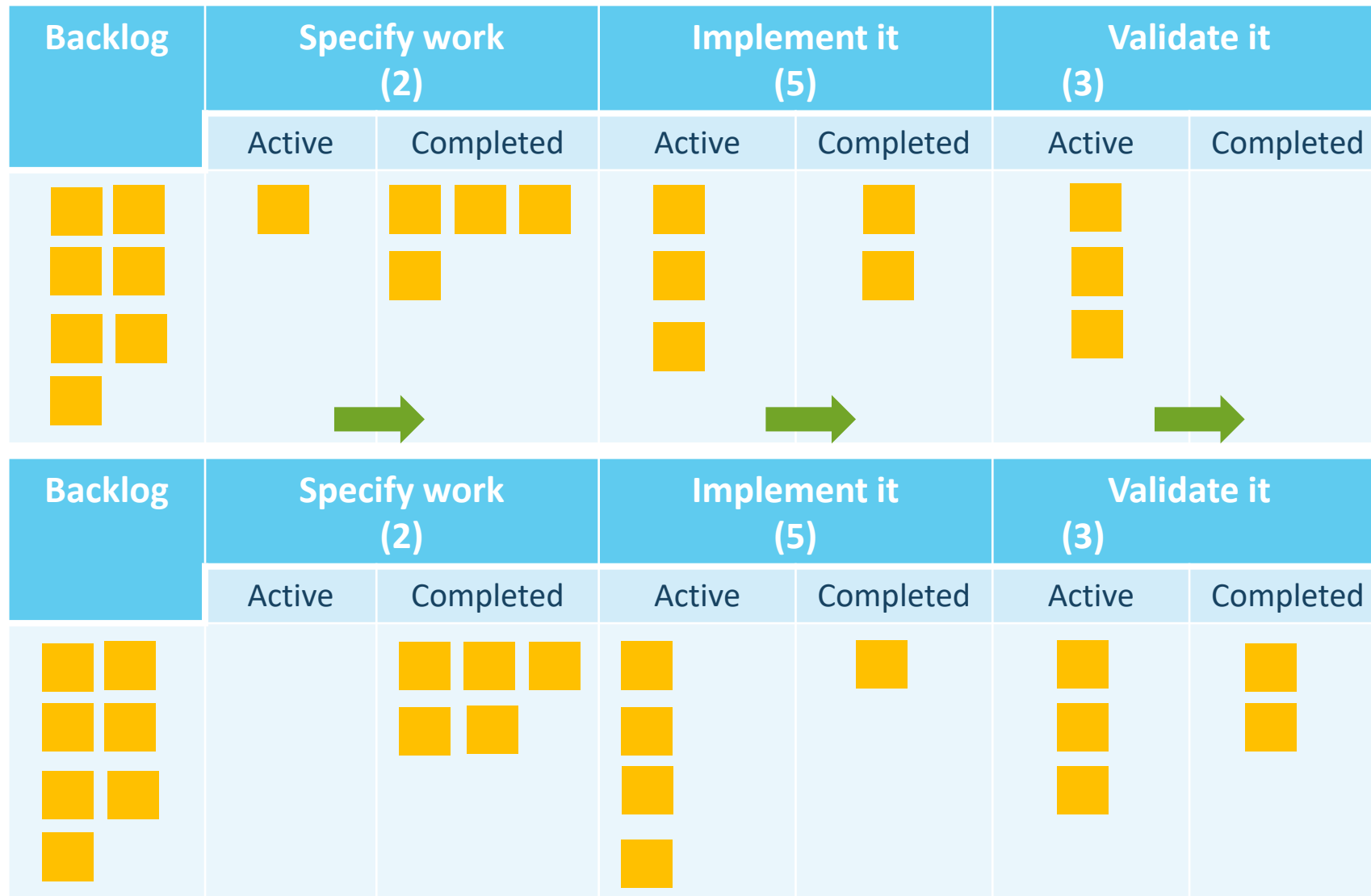
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Kanban dynamics



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Kanban Troubleshooting

- Blocked because all items in an intermediate step are done
- Blocked because prior steps has no items done
- Step taking longer than usual for an item
- Constantly getting blocked
- Item blocked awaiting external input
- Item needs design work
- Plan changes, and updated requirements

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Kanban Checklist

1. Capture your team's high-level routine as a series of steps such as : specify, implement, validate
2. Redecorate your wall with Kanban board that has a Backlog column and two for each of your steps
3. Set limits on chaos or work-in-progress limits (WIP) that restrict the maximum number of work items (cards) for each step.

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Kanban Checklist – cont'd

4. Define “done” rules for each step that must be met before a work item (card) can be moved from Active to Done for that step
5. Run your daily standup at a set time each day, focusing on issues that blocked work from progressing (cards from moving between columns)
6. As needed, update your organization’s tracking system with your team’s current status

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Kanban Checklist – cont'd

7. As needed, adjust your WIP limits and done rules to ensure quality work that flows smoothly through your signboard
8. As needed, add a Track column between the Active and Done columns of your implementation step to track external input that is blocking implementation of a work item (such as a late or an unstable dependency)
9. As needed, add new items (cards) and reorder items in your backlog

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Kanban – Hitting deadlines

- Populate your backlog
- Establish your minimum viable product (MVP)
- Order work, including technical debt (bugs, redesigns, upgrades)
- Estimate features and tasks
- Track expected completion date
- Right-size your team
- Checklist

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Kanban – Hitting deadlines – cont'd

- Establish your minimum viable product (MVP)
 - MVP is usually a small percentage of your backlog, yet once the MVP is done, you'll have a product you can release.
 - MVP avoids unused specifications, incomplete and untested features, unused tests, and unfinished work as it is all essential.
 - Sorting can be done by dividing the backlog into 4 buckets:
 - Must have: MVP - Priority 0
 - Should have - Priority 1
 - Like to have - Priority 2
 - Nice to have - Priority 3

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Kanban – Hitting deadlines – cont'd

- Estimate features and tasks
 - Estimate the number of tasks needed to complete work items
 - Calculate your task completion rate
 - Let your leadership, customers, and internal and external partners know when to expect results

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Kanban – Hitting deadlines – cont'd

- Estimate features and tasks – cont'd
 - How soon will a work item be addressed and done?
Active tasks + estimated tasks for the work item and the work items ahead of it
Task Completion Rate
 - When will a **significant product** release be completed?
Active tasks + estimated tasks for **MVP** work items
Task Completion Rate

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Kanban – Hitting deadlines – cont'd

- Track Expected Completion Date
 1. Task Completion rate = TCR
 2. Current Task Estimate = CTE
 3. Task Add Rate = TAR
 4. Number of days till complete = $CTE / (TCR - TAR)$

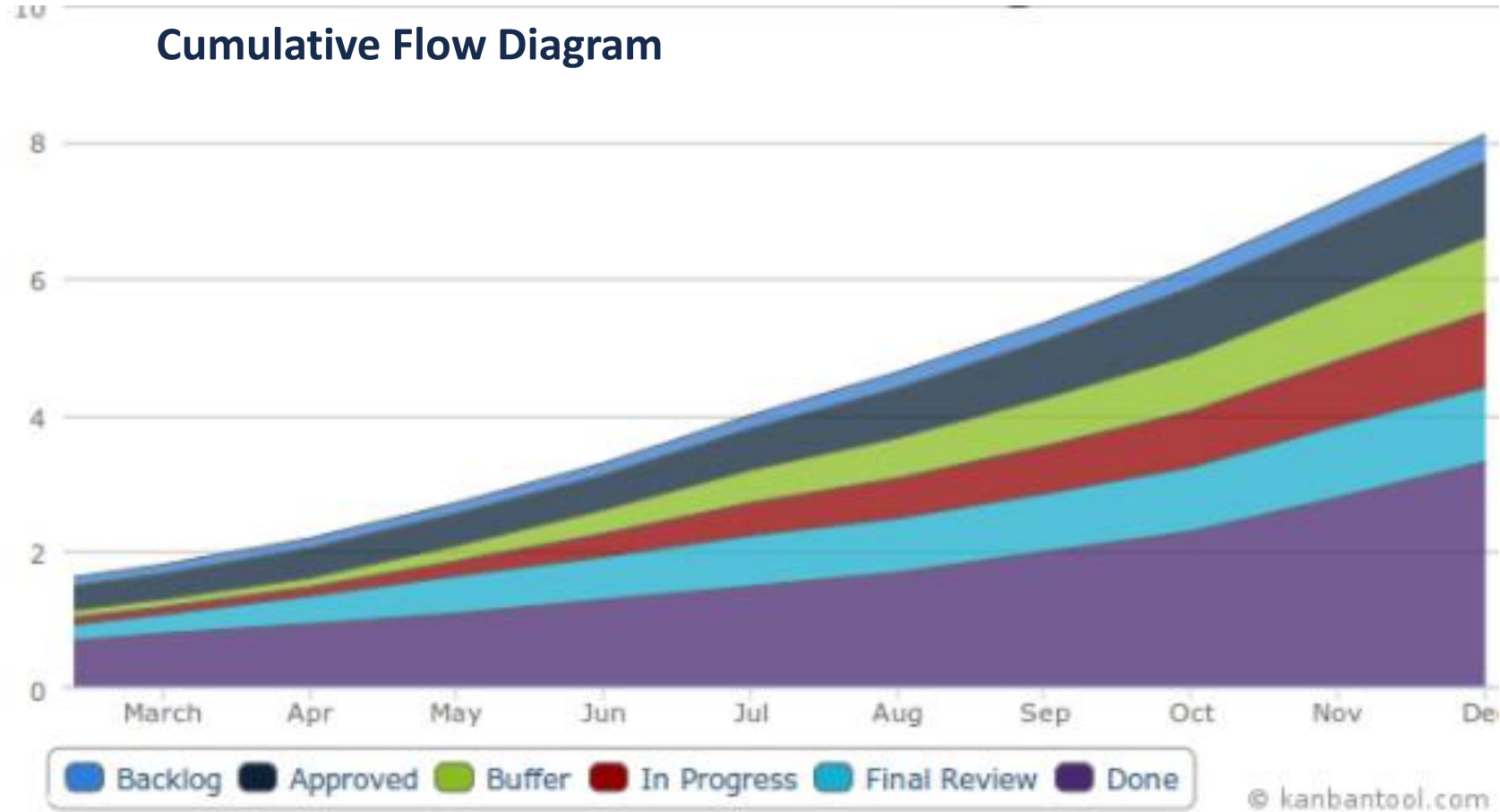
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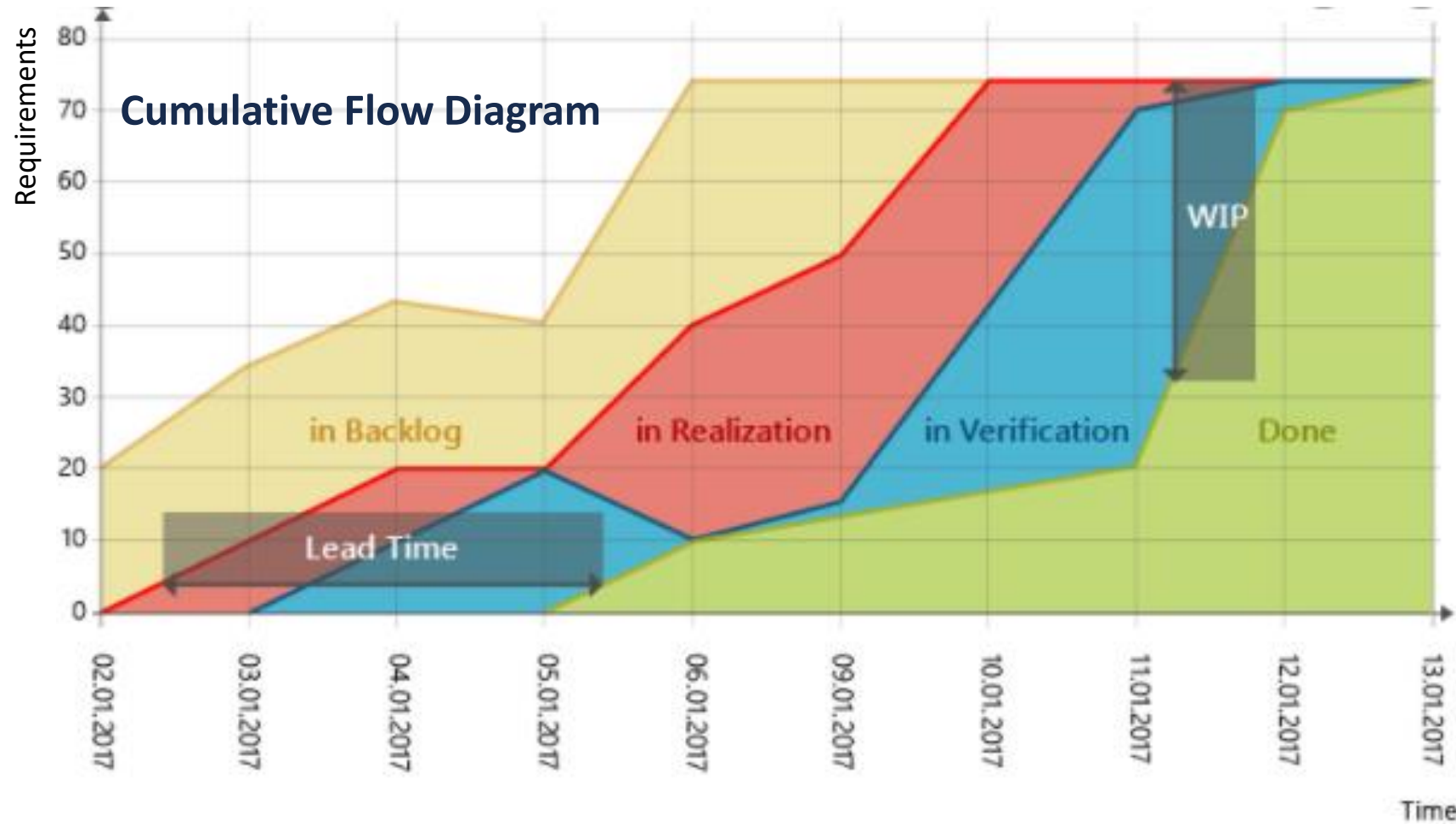
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Kanban – Hitting deadlines – cont'd

- Right-size your team
 - Basic Approach, before your team has measured its CTE, TAR and TCR – use the same data used to determine the initial WIP limits
 - Advanced Approach, after your team has detailed data about the CTE, TAR and TCR – use the data used to track expected completion date

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○ Basic Approach

Step	Specify	Implement	Validate
A: Average rate per month per person	6	2	3
B: Slowest rate (min A:)		2	
C: Estimated people required for step B (items/month/B)		3.07	
D: Throughput of step B (B*C)		6.15	
E: People needed to match B's throughput (D:/A:)	1.02	3.07	2.05
F: WIP limits (E*1.5 rounded up)	2	5	4

on average

constraint

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○ Advanced Approach

CTE	TAR	TCR	CTE/(TCR-TAR) estimate	Target Start Date	Target Completion Date	Days	Estimated /Expected
100	0.10	0.75	152	6/1/2021	10/1/2021	122	1.24

CTE = 100 (4 tasks per basic-example work item)

Step	Specify	Implement	Validate
A: Current WIP limits	2	5	3
B: Current People engaged (A: + 1.5 rounded down)	1	3	2
C: Estimated people required (B: * estimated/expected)	1.24	3.73	2.48
D: Estimated WIP limits (C: * 1.5 rounded up)	2	6	4

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Scrum and Kanban

- *Kanban vs. scrum” is a discussion about two different strategies for implementing an agile development or project management system. Kanban methodologies are continuous and more fluid, whereas scrum is based on short, structured work sprints.”*

Max Rehkopf – Atlassian

- *Don't ask, "Kanban vs Scrum." Instead, ask "Kanban or scrum" or even "Kanban and scrum." Make it more about the principles than the practices.*

<https://www.atlassian.com/agile/kanban/kanban-vs-scrum>

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Scrumban – evolving from Scrum

- Kanban involves every team member through the use of the signboard, note cards, WIP limits, and some rules.
- There is a continuous sprint and only one backlog to manage
- Every note card is an increment on the delivered value
- The note cards in the signboard represent user stories.
- Each large work item can be broken down into smaller tasks

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Scrumban – evolving from Scrum – cont'd

- Team members move the note cards themselves, keeping progress up to date and visible to the team and to customers
- The WIP limit expose problems immediately
- Kanban used done rules at each step to ensure high-quality, usable product always
- Scrum master and Product Owner duties can be divided among the team members over time. You can choose how to manage the roles on your team.

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Scrumban – evolving from Scrum – cont'd

- You can engage leadership and partners separately from customers.
- You can involve more customers than a single Product Owner.
- You can focus individuals on service delivery or other specific needs of your team.
- You can leave your roles exactly as they are under Scrum.

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Scrumban – evolving from Scrum – cont'd

Scrum event	Description	Frequency	Kanban replacement	frequency
High-level planning	PO establishes the initial Product Backlog	Once a project	High-level planning	Once a project
Sprint planning	PO collaborates with the Dev Team to select the highest priority items for the upcoming sprint	At the beginning of each sprint	No event – items pulled from the backlog in order as needed	N/A
Daily scrum	Team answers what was done, what will do before the next daily scrum, what impediments are	Daily	Stand-up meetings but only to the identify impediments	Daily
Sprint review	The Scrum team presents the completed, working software to stakeholders for feedback	At the end of each sprint	No event - customers see the latest product daily or whenever they want	N/A
Retrospective	The team discusses process improvements for next sprint	At the end of each sprint	No event – flow issues are apparent and fixed daily	N/A

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Kanban	Scrum
It is all about visualizing your work, limiting work in progress, and maximizing efficiency(or flow). Kanban teams focus on reducing the time it takes to take a project(or user story) from start to finish. They do this by using a Kanban board and continuously improving their flow of work.	Scrum teams commit to ship working software through set intervals called sprints. Their goal is to create learning loops to quickly gather and integrate customer feedback. Scrum teams adopt specific roles, create special artifacts, and hold regular ceremonies to keep things moving forward. Scrum is best defined in The Scrum Guide.
Agile	
Structured and iterative approach to project management and product development. It recognizes the volatility of product development and provides a methodology for self-organizing teams to respond to change without going off the rails. Today, agile is hardly a competitive advantage. No one has the luxury to develop a product for years or even months in a black box. This means it's more important than ever to get it right.	

<https://www.atlassian.com/agile/kanban/kanban-vs-scrum>

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- 1 - "The Epic Guide to Agile – More Business Value On A Predictable Schedule With Scrum" – Author: Dave Todaro - Copyright © 2019 David M. Todaro – Published by R9 Publishing LLC
- 2 - "Agile Project Management with Scrum" – Author: Ken Schwaber – Copyright © 2004 by Ken Schwaber – Published by Microsoft Press
- 3 - "Agile Project Management with Kanban" – Author: Eric Brechner – Copyright © 2015 by Eric Brechner – Published by Microsoft Press

	Scrum	Kanban
Cadence	Regular fixed length sprints (ie, 2 weeks)	Continuous flow
Release methodology	At the end of each sprint	Continuous delivery
Roles	Product owner, scrum master, development team	No required roles
Key metrics	Velocity	Lead time, cycle time, WIP
Change philosophy	Teams should not make changes during the sprint.	Change can happen at any time

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End of Week 2 / Day 1

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